A Künneth Formula

One might guess that there should be some connection between cup product and product spaces, and indeed this is the case, as we will show in this subsection.

To begin, we define the **cross product**, or **external cup product** as it is sometimes called. This is the map

$$H^*(X;R)\times H^*(Y;R)\xrightarrow{\ \times\ }H^*(X\times Y;R)$$

given by $a\times b=p_1^*(a)\smile p_2^*(b)$ where p_1 and p_2 are the projections of $X\times Y$ onto X and Y. Since cup product is distributive, the cross product is bilinear, that is, linear in each variable separately. We might hope that the cross product map would be an isomorphism in many cases, thereby giving a nice description of the cohomology rings of these product spaces. However, a bilinear map is rarely a homomorphism, so it could hardly be an isomorphism. Fortunately there is a nice algebraic solution to this problem, and that is to replace the direct product $H^*(X;R)\times H^*(Y;R)$ by the tensor product $H^*(X;R)\otimes_R H^*(Y;R)$.

Let us review the definition and basic properties of tensor product. For abelian groups A and B the tensor product $A\otimes B$ is defined to be the abelian group with generators $a\otimes b$ for $a\in A$, $b\in B$, and relations $(a+a')\otimes b=a\otimes b+a'\otimes b$ and $a\otimes(b+b')=a\otimes b+a\otimes b'$. So the zero element of $A\otimes B$ is $0\otimes 0=0\otimes b=a\otimes 0$, and $-(a\otimes b)=-a\otimes b=a\otimes(-b)$. Some readily verified elementary properties are:

(1) $A\otimes B\approx B\otimes A$.
(2) $(\bigoplus_i A_i)\otimes B\approx\bigoplus_i(A_i\otimes B)$.
(3) $(A\otimes B)\otimes C\approx A\otimes(B\otimes C)$.
(4) $\mathbb{Z}\otimes A\approx A$.
(5) $\mathbb{Z}_n\otimes A\approx A/nA$.
(6) A pair of homomorphisms $f:A\to A'$ and $g:B\to B'$ induces a homomorphism $f\otimes g:A\otimes B\to A'\otimes B'$ via $(f\otimes g)(a\otimes b)=f(a)\otimes g(b)$.
(7) A bilinear map $\varphi:A\times B\to C$ induces a homomorphism $A\otimes B\to C$ sending $a\otimes b$ to $\varphi(a,b)$.

In (1)–(5) the isomorphisms are the obvious ones, for example $a\otimes b\mapsto b\otimes a$ in (1) and $n\otimes a\mapsto na$ in (4). Properties (1), (2), (4), and (5) allow the calculation of tensor products of finitely generated abelian groups.

The generalization to tensor product of modules over a commutative ring R is easy. One defines $A\otimes_R B$ for R-modules A and B to be the quotient of $A\otimes B$ obtained by imposing the further relations $ra\otimes b=a\otimes rb$ for $r\in R$, $a\in A$, and $b\in B$. This relation guarantees that $A\otimes_R B$ is again an R-module. In case R is not commutative, one assumes A is a right R-module and B is a left R-module, and the relation is written instead $ar\otimes b=a\otimes rb$, but now $A\otimes_R B$ is only an abelian group, not an R-module necessarily. It is an easy algebra exercise to see that $A\otimes_R B=A\otimes B$ when R

is $\mathbb{Z}_m$ or $\mathbb{Q}$. But in general $A\otimes_R B$ is not the same as $A\otimes B$. For example, if $R=\mathbb{Q}(\sqrt{2})$, a 2-dimensional vector space over $\mathbb{Q}$, then $R\otimes_R R=R$ but $R\otimes R$ is a 4-dimensional vector space over $\mathbb{Q}$.

The statements (1)–(3), (6), and (7) remain valid for tensor products of R-modules. The generalization of (4) is the canonical isomorphism $R\otimes_R A\approx A$, $r\otimes a\mapsto ra$.

Property (7) of tensor products guarantees that the cross product as defined above gives rise to a homomorphism

$$H^*(X;R)\otimes_R H^*(Y;R)\xrightarrow{\ \times\ }H^*(X\times Y;R),\qquad a\otimes b\mapsto a\times b$$

which we shall also call cross product. This map becomes a ring homomorphism if we define the multiplication in a tensor product of graded rings by $(a\otimes b)(c\otimes d)=(-1)^{|b||c|}ac\otimes bd$ where $|x|$ denotes the dimension of x. Namely, the cross product map sends $(a\otimes b)(c\otimes d)=(-1)^{|b||c|}ac\otimes bd$ to

$$\begin{aligned}(-1)^{|b||c|}ac\times bd&=(-1)^{|b||c|}p_1^*(a\smile c)\smile p_2^*(b\smile d)\\&=(-1)^{|b||c|}p_1^*(a)\smile p_1^*(c)\smile p_2^*(b)\smile p_2^*(d)\\&=p_1^*(a)\smile p_2^*(b)\smile p_1^*(c)\smile p_2^*(d)\\&=(a\times b)(c\times d)\end{aligned}$$

which is the product of the images of $a\otimes b$ and $c\otimes d$.

Theorem 3.16. *The cross product $H^*(X;R)\otimes_R H^*(Y;R)\to H^*(X\times Y;R)$ is an isomorphism of rings if X and Y are CW complexes and $H^k(Y;R)$ is a finitely generated free R-module for all k.*

Results of this type, computing homology or cohomology of a product space, are known as **Künneth formulas**. The hypothesis that X and Y are CW complexes will be shown to be unnecessary in §4.1 when we consider CW approximations to arbitrary spaces. On the other hand, the freeness hypothesis cannot always be dispensed with, as we shall see in §3.B when we obtain a completely general Künneth formula for the homology of a product space.

When the conclusion of the theorem holds, the ring structure in $H^*(X\times Y;R)$ is determined by the ring structures in $H^*(X;R)$ and $H^*(Y;R)$. Example 3E.6 shows that some hypotheses are necessary in order for this to be true.

Before proving the theorem, let us look at some examples.

Example 3.17. The theorem says that $H^*(\mathbb{R}\mathrm{P}^\infty\times\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2)$ is isomorphic as a ring to $H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2)\otimes H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2)$. By Theorem 3.12 this is $\mathbb{Z}_2[\alpha]\otimes\mathbb{Z}_2[\beta]$, which is just the polynomial ring $\mathbb{Z}_2[\alpha,\beta]$. More generally we see by induction that for a product of n copies of $\mathbb{R}\mathrm{P}^\infty$, the $\mathbb{Z}_2$-cohomology is a polynomial ring in n variables. Similar remarks apply to $\mathbb{C}\mathrm{P}^\infty$ and $\mathbb{H}\mathrm{P}^\infty$ with coefficients in an arbitrary commutative ring.

Example 3.18. The exterior algebra $\Lambda_R[\alpha_1,\alpha_2,\cdots]$ is the graded tensor product over R of the one-variable exterior algebras $\Lambda_R[\alpha_i]$ where the α_i's have odd dimension. The Künneth formula then gives an isomorphism $H^*(S^{k_1}\times\cdots\times S^{k_n};\mathbb{Z})\approx\Lambda_{\mathbb{Z}}[\alpha_1,\cdots,\alpha_n]$ if the dimensions k_i are all odd. With some k_i's even, one would have the tensor product of an exterior algebra for the odd-dimensional spheres and truncated polynomial rings $\mathbb{Z}[\alpha]/(\alpha^2)$ for the even-dimensional spheres. Of course, $\Lambda_{\mathbb{Z}}[\alpha]$ and $\mathbb{Z}[\alpha]/(\alpha^2)$ are isomorphic as rings, but when one takes tensor products in the graded sense it becomes important to distinguish them as graded rings, with α odd-dimensional in $\Lambda_{\mathbb{Z}}[\alpha]$ and even-dimensional in $\mathbb{Z}[\alpha]/(\alpha^2)$. These remarks apply more generally with any coefficient ring R in place of $\mathbb{Z}$, though when $R=\mathbb{Z}_2$ there is no need to distinguish between the odd-dimensional and even-dimensional cases since signs become irrelevant.

The idea of the proof of the theorem will be to consider, for a fixed CW complex Y, the functors

$$h^n(X,A)=\bigoplus_i\big(H^i(X,A;R)\otimes_R H^{n-i}(Y;R)\big)$$
$$k^n(X,A)=H^n(X\times Y,A\times Y;R)$$

The cross product, or a relative version of it, defines a map $\mu:h^n(X,A)\to k^n(X,A)$ which we want to show is an isomorphism when X is a CW complex and $A=\varnothing$. We will show:

(1) h^* and k^* are cohomology theories on the category of CW pairs.

(2) μ is a natural transformation: It commutes with induced homomorphisms and with coboundary homomorphisms in long exact sequences of pairs.

It is obvious that $\mu:h^n(X)\to k^n(X)$ is an isomorphism when X is a point since it is just the scalar multiplication map $R\otimes_R H^n(Y;R)\to H^n(Y;R)$. The following general fact will then imply the theorem.

Proposition 3.19. *If a natural transformation between unreduced cohomology theories on the category of CW pairs is an isomorphism when the CW pair is* $(point,\varnothing)$*, then it is an isomorphism for all CW pairs.*

Proof: Let $\mu:h^*(X,A)\to k^*(X,A)$ be the natural transformation. By the five-lemma it will suffice to show that μ is an isomorphism when $A=\varnothing$.

First we do the case of finite-dimensional X by induction on dimension. The induction starts with the case that X is 0-dimensional, where the result holds by hypothesis and by the axiom for disjoint unions. For the induction step, μ gives a map between the two long exact sequences for the pair (X^n,X^{n-1}), with commuting squares since μ is a natural transformation. The five-lemma reduces the inductive step to showing that μ is an isomorphism for $(X,A)=(X^n,X^{n-1})$. Let $\Phi:\coprod_\alpha(D^n_\alpha,\partial D^n_\alpha)\to(X^n,X^{n-1})$ be a collection of characteristic maps for all the n-cells of X. By excision, Φ^* is an isomorphism for h^* and k^*, so by naturality it suffices to show that μ is an isomorphism for $(X,A) = \coprod_\alpha (D^n_\alpha, \partial D^n_\alpha)$. The axiom for disjoint unions gives a further reduction to the case of the pair $(D^n, \partial D^n)$. Finally, this case follows by applying the five-lemma to the long exact sequences of this pair, since D^n is contractible and hence is covered by the 0-dimensional case, and ∂D^n is $(n-1)$-dimensional.

The case that X is infinite-dimensional reduces to the finite-dimensional case by a telescope argument as in the proof of Lemma 2.34. We leave this for the reader since the finite-dimensional case suffices for the special h^* and k^* we are considering, as $h^i(X,X^n)$ and $k^i(X,X^n)$ are both zero when $n \ge i$, by cellular cohomology for example. $\square$

Proof of 3.16: It remains to check that h^* and k^* are cohomology theories, and that μ is a natural transformation. Since we are dealing with unreduced cohomology theories there are four axioms to verify.

(1) Homotopy invariance: $f \simeq g$ implies $f^* = g^*$. This is obvious for both h^* and k^*.

(2) Excision: $h^*(X,A) \approx h^*(B, A\cap B)$ for A and B subcomplexes of the CW complex $X = A \cup B$. This is obvious, and so is the corresponding statement for k^* since $(A\times Y) \cup (B\times Y) = (A\cup B)\times Y$ and $(A\times Y)\cap(B\times Y) = (A\cap B)\times Y$.

(3) The long exact sequence of a pair. This is a triviality for k^*, but a few words of explanation are needed for h^*, where the desired exact sequence is obtained in two steps. For the first step, tensor the long exact sequence of ordinary cohomology groups for a pair (X,A) with the free R-module $H^n(Y;R)$, for a fixed n. This yields another exact sequence because $H^n(Y;R)$ is a direct sum of copies of R, so the result of tensoring an exact sequence with this direct sum is simply to produce a direct sum of copies of the exact sequence, which is again an exact sequence. The second step is to let n vary, taking a direct sum of the previously constructed exact sequences for each n, with the n^{th} exact sequence shifted up by n dimensions.

(4) Disjoint unions. Again this axiom obviously holds for k^*, but some justification is required for h^*. What is needed is the algebraic fact that there is a canonical isomorphism $(\prod_\alpha M_\alpha)\otimes_R N \approx \prod_\alpha (M_\alpha \otimes_R N)$ for R-modules M_α and a finitely generated free R-module N. Since N is a direct product of finitely many copies R_β of R, $M_\alpha \otimes_R N$ is a direct product of corresponding copies $M_{\alpha\beta} = M_\alpha \otimes_R R_\beta$ of M_α and the desired relation becomes $\prod_\beta \prod_\alpha M_{\alpha\beta} \approx \prod_\alpha \prod_\beta M_{\alpha\beta}$, which is obviously true.

Finally there is naturality of μ to consider. Naturality with respect to maps between spaces is immediate from the naturality of cup products. Naturality with respect to coboundary maps in long exact sequences is commutativity of the square displayed in Example 3.11. $\square$

The following theorem of Hopf is a nice algebraic application of the cup product structure in $H^*(\mathbb{R}\mathrm{P}^n \times \mathbb{R}\mathrm{P}^n; \mathbb{Z}_2)$ described by the Künneth formula.

Theorem 3.20. *If $\mathbb{R}^n$ has the structure of a division algebra over the scalar field $\mathbb{R}$, then n must be a power of 2.*

Proof: Given a division algebra structure on $\mathbb{R}^n$, define a map $g : S^{n-1} \times S^{n-1} \to S^{n-1}$ by $g(x,y) = xy/|xy|$. This is well-defined since there are no zero divisors, and continuous by the bilinearity of the multiplication. From the relations $(-x)y = -(xy) = x(-y)$ it follows that $g(-x,y) = -g(x,y) = g(x,-y)$. This implies that g induces a quotient map $h : \mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1} \to \mathbb{R}\mathrm{P}^{n-1}$.

We claim that $h^* : H^1(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{Z}_2) \to H^1(\mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1}; \mathbb{Z}_2)$ is the map $h^*(\gamma) = \alpha + \beta$ where γ generates $H^1(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{Z}_2)$ and α and β are the pullbacks of γ under the projections of $\mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1}$ onto its two factors. This can be proved as follows. We may assume $n > 2$, so $\pi_1(\mathbb{R}\mathrm{P}^{n-1}) \approx \mathbb{Z}_2$. Let $\lambda : I \to S^{n-1}$ be a path joining a point x to the antipodal point $-x$. Then for fixed y, the path $s \mapsto g(\lambda(s), y)$ joins $g(x,y)$ to $g(-x,y) = -g(x,y)$. Hence, identifying antipodal points, h takes a nontrivial loop in the first $\mathbb{R}\mathrm{P}^{n-1}$ factor of $\mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1}$ to a nontrivial loop in $\mathbb{R}\mathrm{P}^{n-1}$. The same argument works for the second factor, so the restriction of h to the 1-skeleton $S^1 \vee S^1$ is homotopic to the map that includes each S^1 summand of $S^1 \vee S^1$ into $\mathbb{R}\mathrm{P}^{n-1}$ as the 1-skeleton. Since restriction to the 1-skeleton is an isomorphism on $H^1(-; \mathbb{Z}_2)$ for both $\mathbb{R}\mathrm{P}^{n-1}$ and $\mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1}$, it follows that $h^*(\gamma) = \alpha + \beta$.

Since $\gamma^n = 0$ we have $0 = h^*(\gamma^n) = (\alpha+\beta)^n = \sum_k \binom{n}{k} \alpha^k \beta^{n-k}$. This is an equation in the ring $H^*(\mathbb{R}\mathrm{P}^{n-1} \times \mathbb{R}\mathrm{P}^{n-1}; \mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha,\beta]/(\alpha^n, \beta^n)$, so the coefficient $\binom{n}{k}$ must be zero in $\mathbb{Z}_2$ for all k in the range $0 < k < n$. It is a rather easy number theory fact that this happens only when n is a power of 2. Namely, an obviously equivalent statement is that in the polynomial ring $\mathbb{Z}_2[x]$, the equality $(1+x)^n = 1 + x^n$ holds only when n is a power of 2. To prove the latter statement, write n as a sum of powers of 2, $n = n_1 + \cdots + n_k$ with $n_1 < \cdots < n_k$. Then $(1+x)^n = (1+x)^{n_1} \cdots (1+x)^{n_k} = (1+x^{n_1}) \cdots (1+x^{n_k})$ since squaring is an additive homomorphism with $\mathbb{Z}_2$ coefficients. If one multiplies the product $(1+x^{n_1}) \cdots (1+x^{n_k})$ out, no terms combine or cancel since $n_i \ge 2n_{i-1}$ for each i, and so the resulting polynomial has 2^k terms. Thus if this polynomial equals $1 + x^n$ we must have $k = 1$, which means that n is a power of 2. $\square$

It is sometimes important to have a relative version of the Künneth formula in Theorem 3.16. The relative cross product is

$$H^*(X,A;R) \otimes_R H^*(Y,B;R) \xrightarrow{\times} H^*(X \times Y, A \times Y \cup X \times B; R)$$

for CW pairs (X,A) and (Y,B), defined just as in the absolute case by $a \times b = p_1^*(a) \smile p_2^*(b)$ where $p_1^*(a) \in H^*(X \times Y, A \times Y; R)$ and $p_2^*(b) \in H^*(X \times Y, X \times B; R)$.

Theorem 3.21. *For CW pairs (X,A) and (Y,B) the cross product homomorphism $H^*(X,A;R)\otimes_R H^*(Y,B;R)\to H^*(X\times Y, A\times Y\cup X\times B;R)$ is an isomorphism of rings if $H^k(Y,B;R)$ is a finitely generated free R-module for each k.*

Proof: The case $B=\varnothing$ was covered in the course of the proof of the absolute case, so it suffices to deduce the case $B\neq\varnothing$ from the case $B=\varnothing$.

The following commutative diagram shows that collapsing B to a point reduces the proof to the case that B is a point:

$$\begin{tikzcd}
H^*(X,A)\otimes_R H^*(Y,B) \arrow[d, "\times"] & H^*(X,A)\otimes_R H^*(Y/B,B/B) \arrow[l, "\approx"'] \arrow[d, "\times"] \\
H^*(X\times Y, A\times Y\cup X\times B) & H^*(X\times(Y/B), A\times(Y/B)\cup X\times(B/B)) \arrow[l, "\approx"']
\end{tikzcd}$$

The lower map is an isomorphism since the quotient spaces $(X\times Y)/(A\times Y\cup X\times B)$ and $(X\times(Y/B))/(A\times(Y/B)\cup X\times(B/B))$ are the same.

In the case that B is a point $y_0\in Y$, consider the commutative diagram

$$\begin{tikzcd}
H^*(X,A)\otimes_R H^*(Y,y_0) \arrow[r] \arrow[dd, "\times"] & H^*(X,A)\otimes_R H^*(Y) \arrow[r] \arrow[dd, "\times"] & H^*(X,A)\otimes_R H^*(y_0) \arrow[d, "\times"] \\
& & H^*(X\times y_0, A\times y_0) \\
H^*(X\times Y, X\times y_0\cup A\times Y) \arrow[r] & H^*(X\times Y, A\times Y) \arrow[r] \arrow[ur] & H^*(X\times y_0\cup A\times Y, A\times Y) \arrow[u, "\approx"']
\end{tikzcd}$$

Since y_0 is a retract of Y, the upper row of this diagram is a split short exact sequence. The lower row is the long exact sequence of a triple, and it too is a split short exact sequence since $(X\times y_0, A\times y_0)$ is a retract of $(X\times Y, A\times Y)$. The middle and right cross product maps are isomorphisms by the case $B=\varnothing$ since $H^k(Y;R)$ is a finitely generated free R-module if $H^k(Y,y_0;R)$ is. The five-lemma then implies that the left-hand cross product map is an isomorphism as well. ◻

The relative cross product for pairs (X,x_0) and (Y,y_0) gives a reduced cross product

$$\widetilde{H}^*(X;R)\otimes_R\widetilde{H}^*(Y;R)\xrightarrow{\ \times\ }\widetilde{H}^*(X\wedge Y;R)$$

where $X\wedge Y$ is the smash product $X\times Y/(X\times\{y_0\}\cup\{x_0\}\times Y)$. The preceding theorem implies that this reduced cross product is an isomorphism if $\widetilde{H}^*(X;R)$ or $\widetilde{H}^*(Y;R)$ is free and finitely generated in each dimension. For example, we have isomorphisms $\widetilde{H}^n(X;R)\approx\widetilde{H}^{n+k}(X\wedge S^k;R)$ via cross product with a generator of $H^k(S^k;R)\approx R$. The space $X\wedge S^k$ is the k-fold reduced suspension $\Sigma^k X$ of X, so we see that the suspension isomorphisms $\widetilde{H}^n(X;R)\approx\widetilde{H}^{n+k}(\Sigma^k X;R)$ derivable by elementary exact sequence arguments can also be obtained via cross product with a generator of $\widetilde{H}^*(S^k;R)$.

Spaces with Polynomial Cohomology

We saw in Theorem 3.12 that $\mathbb{R}P^\infty$, $\mathbb{C}P^\infty$, and $\mathbb{H}P^\infty$ have cohomology rings that are polynomial algebras. We will describe now a construction for enlarging S^{2n} to a space $J(S^{2n})$ whose cohomology ring $H^*(J(S^{2n});\mathbb{Z})$ is almost the polynomial ring $\mathbb{Z}[x]$ on a generator x of dimension $2n$. And if we change from $\mathbb{Z}$ to $\mathbb{Q}$ coefficients, then $H^*(J(S^{2n});\mathbb{Q})$ is exactly the polynomial ring $\mathbb{Q}[x]$. This construction, known as the **James reduced product**, is also of interest because of its connections with loopspaces described in §4.J.

For a space X, let X^k be the product of k copies of X. From the disjoint union $\coprod_{k\ge1} X^k$, let us form a quotient space $J(X)$ by identifying $(x_1,\cdots,x_i,\cdots,x_k)$ with $(x_1,\cdots,\widehat{x}_i,\cdots,x_k)$ if $x_i = e$, a chosen basepoint of X. Points of $J(X)$ can thus be thought of as k-tuples $(x_1,\cdots,x_k)$, $k\ge 0$, with no $x_i = e$. Inside $J(X)$ is the subspace $J_m(X)$ consisting of the points $(x_1,\cdots,x_k)$ with $k\le m$. This can be viewed as a quotient space of X^m under the identifications $(x_1,\cdots,x_i,e,\cdots,x_m)\sim(x_1,\cdots,e,x_i,\cdots,x_m)$. For example, $J_1(X) = X$ and $J_2(X) = X\times X/(x,e)\sim(e,x)$. If X is a CW complex with e a 0-cell, the quotient map $X^m \to J_m(X)$ glues together the m subcomplexes of the product complex X^m where one coordinate is e. These glueings are by homeomorphisms taking cells onto cells, so $J_m(X)$ inherits a CW structure from X^m. There are natural inclusions $J_m(X)\subset J_{m+1}(X)$ as subcomplexes, and $J(X)$ is the union of these subcomplexes, hence is also a CW complex.

Proposition 3.22. *For $n>0$, $H^*(J(S^n);\mathbb{Z})$ consists of a $\mathbb{Z}$ in each dimension a multiple of n. If n is even, the i^{th} power of a generator of $H^n(J(S^n);\mathbb{Z})$ is $i!$ times a generator of $H^{in}(J(S^n);\mathbb{Z})$, for each $i\ge 1$.*

Thus for n even, $H^*(J(S^n);\mathbb{Z})$ can be identified with the subring of the polynomial ring $\mathbb{Q}[x]$ additively generated by the monomials $x^i/i!$. This subring is called a **divided polynomial algebra** and is denoted $\Gamma_{\mathbb{Z}}[x]$. An exercise at the end of the section is to show that when n is odd, $H^*(J(S^n);\mathbb{Z})$ is isomorphic as a graded ring to $H^*(S^n;\mathbb{Z})\otimes H^*(J(S^{2n});\mathbb{Z})$, the tensor product of an exterior algebra and a divided polynomial algebra.

Proof: Giving S^n its usual CW structure, the resulting CW structure on $J(S^n)$ consists of exactly one cell in each dimension a multiple of n. Thus if $n>1$ we deduce immediately from cellular cohomology that $H^*(J(S^n);\mathbb{Z})$ consists exactly of $\mathbb{Z}$'s in dimensions a multiple of n. An alternative argument that works also when $n=1$ is the following. Consider the quotient map $q:(S^n)^m \to J_m(S^n)$. This carries each cell of $(S^n)^m$ homeomorphically onto a cell of $J_m(S^n)$. In particular q is a cellular map, taking k-skeleton to k-skeleton for each k, so q induces a chain map of cellular chain complexes. This chain map is surjective since each cell of $J_m(S^n)$ is the homeomorphic image of a cell of $(S^n)^m$. Hence all the cellular boundary maps for $J_m(S^n)$